Student Challenge: Smart FabLab Access Control Terminal

Overview

Design, program, and fabricate a wall-mounted NFC access terminal powered by an ESP32 WROOM-32 microcontroller. The device will scan member NFC tags, query a central Google Sheet to retrieve safety and machine permissions, and display access rights across multiple navigable pages on a local LCD screen.

Hardware Requirements

- **Microcontroller:** ESP32 WROOM-32 Development Board
- **NFC Module:** RFID-RC522 (13.56 MHz) SPI module
- **Display:** Standard 16x2 LCD (with I2C Backpack module, default address 0x27 or 0x3F)
- **User Input:** At least 1 physical push button (configured with internal software pull-up/pull-down)
- **Provided Test Materials:** 3x assigned NFC tags

Technical & Functional Specifications

1. Database & Network Integration

- Connect the ESP32 to local Wi-Fi.
- Upon scanning an NFC tag, send an HTTP request to the provided public Google Apps Script endpoint mapped to the central Google Sheet.
- Retrieve and parse the JSON payload containing member permissions:
  - FabLab Onboarding (TRUE / FALSE)
  - Laser Cutter Access (TRUE / FALSE)
  - CNC Access (TRUE / FALSE)
  - Bambu H2C Access (TRUE / FALSE)

2. Display Navigation & Pagination

The physical push button must cycle through information pages on the 16x2 LCD screen upon pressing:

- **Page 0 (Default / Registration View):**
  - Displays the raw hexadecimal NFC Tag UID (e.g., UID: A1:B2:C3:D4).
  - Used to read and register new tags during the onboarding process.
- **Page 1 (Member Profile & General Access):**
  - Displays the Member's Name and FabLab Onboarding status (e.g., Jane Doe | Onboarded: YES).
- **Page 2 (Heavy Machinery Rights):**
  - Displays status for Laser Cutter and CNC Access (e.g., Laser: OK | CNC: NO).
- **Page 3 (3D Printing Rights):**
  - Displays status for Bambu H2C Access (e.g., Bambu H2C: OK).

Requirement: The display must automatically reset to an idle "Tap Tag to Scan" state

after 15 seconds of inactivity.

3. Engineering Challenge: First-Scan Name Association

When a brand-new, unassigned NFC tag is swiped, its UID will not have a corresponding member name in the database.

- **Your Task:** Propose and implement a practical mechanism to associate a person's name with their NFC Tag UID upon its **first scan**.
- **Suggested Approaches to Explore:**
  - **Auto-Queueing:** The ESP32 sends unknown UIDs to a "Unassigned Tags" sheet tab, triggering a pre-filled Google Form link for administrative onboarding.
  - **Local Configuration Webserver:** The ESP32 hosts a temporary local web portal where an instructor can type in a member's name upon scanning an unrecognized tag.
  - **Dynamic Sheet Creation:** Google Apps Script automatically appends a new row with the scanned UID and a placeholder name ("NEW USER - EDIT NAME"), allowing an admin to complete the entry later.

Physical Enclosure & Electronics Specs

- **Wall-Mountable CAD Case:**
  - Design a custom 3D-printable enclosure featuring keyhole slots or external mounting tabs for wall attachment.
  - Include precise cutouts for the 16x2 LCD screen, the push button, and a visual target indicating where to tap the NFC tag.
- **Permanent Power System:**
  - Implement a permanent power delivery solution (e.g., hardwired 5V USB wall adapter, recessed USB-C pass-through connector, or internal step-down buck converter) rather than relying on temporary breadboard/bench power.

Deliverables & Evaluation

1. **Functional Hardware Prototype:** Fully wired and stable ESP32 access control unit.
2. **Working Firmware:** C++/Arduino code handling Wi-Fi reconnects, SPI/I2C communication, JSON parsing, and button interrupts/debouncing.
3. **3D CAD Files:** Print-ready .STL / .3MF files and source parametric models for the enclosure.
4. **Architecture Summary (1 Page):** Short write-up detailing your team's solution for the "First-Scan Name Association" challenge and power delivery design choices.